Explicit disjoint zero blocks for square-root floor differences

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Abstract

We prove the disjoint zero-block conjecture of Chamberland and Dilcher. An explicit formula places each block inside the finite interval required by their alternating-sum problem. The blocks are disjoint and partition all zero positions. The proof is a bijection obtained by reflecting two consecutive integers across $n/2$.

Task 1: the original conjecture and its resolution

Fix an odd positive integer n, and put $N = (n-1)/2$. Following [1, Equation (2.3) and Conjecture 2.1], set

$$d_n(\ell) = \left\lfloor \sqrt{2\ell n} \right\rfloor - \left\lfloor \sqrt{(2\ell-1)n} \right\rfloor, \qquad 1 \le \ell \le N.$$

The source asks that a value $d_n(\lambda) = \delta \ge 2$ be associated with $\delta - 1$ consecutive zero positions, with different associations disjoint. The finite interval is part of the source problem: choosing zeros arbitrarily far beyond N would not answer it.

Theorem 1. *For $1 \le \lambda \le N$ with $\delta = d_n(\lambda) \ge 2$, define*

$$a_\lambda = \left\lfloor \sqrt{(2\lambda-1)n} \right\rfloor, \qquad b_\lambda = \left\lfloor \sqrt{2\lambda n} \right\rfloor, \qquad L_\lambda = \frac{n+1}{2} + \lambda - b_\lambda.$$

Then

$$B_\lambda = \{L_\lambda, L_\lambda + 1, \ldots, L_\lambda + \delta - 2\} \subseteq \{1, \ldots, N\}$$

consists entirely of zeros of d_n. These blocks are pairwise disjoint, and their union is exactly $\{\ell : 1 \le \ell \le N,\ d_n(\ell) = 0\}$. The assertion is vacuous when $n = 1$.

For example, for $n = 33$ the surplus positions are $\lambda = 1, 2, 3$, with values $3, 2, 2$. The formula gives respectively $B_1 = \{10, 11\}$, $B_2 = \{8\}$, and $B_3 = \{6\}$, reproducing the source's example.

Two ceiling identities

For integer $0 \le k \le n$, define $c(k) = \lceil k^2/n \rceil$. Integer translation inside a ceiling gives the exact reflection identity

$$c(n-k) = n - 2k + c(k). \tag{1}$$

Moreover, for $0 \le r \le n-1$,

$$0 \le c(r+1) - c(r) \le 2, \tag{2}$$

since $((r+1)^2 - r^2)/n = (2r+1)/n < 2$. Finally,

$$c(k) = 2\lambda \quad \Longleftrightarrow \quad a_\lambda + 1 \le k \le b_\lambda. \tag{3}$$

The strict lower and weak upper square inequalities in this equivalence also cover the case where an endpoint is a perfect square.

Constructing each block and proving disjointness

Fix λ as in the theorem. For each

$$k \in \{a_\lambda + 2, \dots, b_\lambda\}, \qquad r = n - k, \qquad \ell = N + 1 + \lambda - k,$$

equation (3) gives $c(k-1) = c(k) = 2\lambda$. Applying (1) to these two integers yields

$$c(r) = 2\ell - 1, \qquad c(r+1) = 2\ell + 1. \tag{4}$$

Therefore

$$r^2 \le (2\ell - 1)n \le 2\ell n < (r+1)^2,$$

and hence

$$\left\lfloor \sqrt{(2\ell-1)n} \right\rfloor = \left\lfloor \sqrt{2\ell n} \right\rfloor = r. \tag{5}$$

Thus $d_n(\ell) = 0$.

We also check the interval, rather than assuming it. Since $\lambda \le N$, we have $b_\lambda \le n-1$, so $r \ge 1$. Consequently $c(r) \ge 1$, and (4) implies $\ell \ge 1$. Also $k - 1 \ge 1$, so $r + 1 \le n$ and $c(r+1) \le n$; thus $2\ell + 1 \le n$ and $\ell \le N$.

As k runs consecutively from $a_\lambda + 2$ to b_λ, ℓ runs consecutively in reverse order through exactly B_λ. This proves its size, position, and zero property. If the same ℓ arose from two pairs (λ, k), equation (5) would determine the same r, and hence the same $k = n - r$. The equality $\ell = N + 1 + \lambda - k$ then determines the same λ. This proves disjointness.

The inverse map: every zero occurs

Let $1 \le \ell \le N$ satisfy $d_n(\ell) = 0$, and set

$$r = \left\lfloor \sqrt{2\ell n} \right\rfloor = \left\lfloor \sqrt{(2\ell-1)n} \right\rfloor.$$

Then $r^2 \le (2\ell-1)n$ and $(r+1)^2 > 2\ell n$, so $c(r) \le 2\ell - 1$ and $c(r+1) \ge 2\ell + 1$. Here $1 \le r \le n-1$. The gap bound (2) forces equality:

$$c(r) = 2\ell - 1, \qquad c(r+1) = 2\ell + 1.$$

Set $k = n - r$ and $\lambda = N + \ell - r$. Reflection now gives

$$c(k-1) = c(k) = 2\lambda.$$

In particular $k - 1 > 0$: otherwise the left side would be $c(0) = 0$, while $c(k) > 0$. Since $k \le n - 1$, the common value is a positive even integer at most $n - 1$; thus $1 \le \lambda \le N$. Equation (3) shows $a_\lambda + 2 \le k \le b_\lambda$, so $d_n(\lambda) \ge 2$ and the construction above maps this pair to the original ℓ. This proves surjectivity and the theorem.

The already known alternating-sum identity. The bijection gives

$$\#\{\ell : d_n(\ell) = 0\} = \sum_{\lambda : d_n(\lambda) \ge 2} (d_n(\lambda) - 1), \qquad \sum_{\ell=1}^{N} d_n(\ell) = N.$$

It follows that $\sum_{j=1}^{n} (-1)^{j+1} \left\lfloor \sqrt{jn} \right\rfloor = n - N = (n+1)/2$. This last identity is already Theorem 1.1 of [1]; no new credit is claimed for it. The new target here is the explicit consecutive, disjoint block structure left conjectural in that paper.

Tasks 2–5: follow-up questions

2. Determine an asymptotic formula, with an explicit error term, for the number of zero positions as odd $n \to \infty$.

3. Determine the number of blocks of each length and the maximal block length, including the dependence on square endpoints.

4. Classify the analogous zero and surplus structure for even n, where reflection preserves, rather than reverses, ceiling parity.

5. Identify higher-power floor differences admitting a comparable block bijection, and determine which reflection or increment conditions are necessary for such a construction.

Verification and scope. The proof covers every odd positive integer, including composite integers and square endpoints. The accompanying standard-library verifier uses integer square roots and integer division only. Its finite exhaustive and large-integer checks corroborate the proof; they do not replace it. The source and prior-claim searches are bounded, and their limits are recorded in the literature audit. No independent peer review is claimed.

References

[1] M. Chamberland and K. Dilcher, *An alternating sum of the floor function of square roots*, arXiv:2510.26291v1 (2025), Conjecture 2.1, pp. 2–3. Complete current five-page manuscript checked 29 September 2026.

[2] MathDB, problem 371228, live status and activity checked 29 September 2026.